

SI 140A-01 Probability & Statistics for EECS, Fall 2026

Homework 2

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Read all the instructions below carefully before you start working on the assignment, and before you make a submission.

- You are required to write down all the major steps towards making your conclusions; otherwise you may obtain limited points of the problem.
- Write your homework in English; otherwise you will get no points of this homework.
- Any form of plagiarism will lead to 0 point of this homework.

Problem 1

Show that for any events A and B ,

$$P(A) + P(B) - 1 \leq P(A \cap B) \leq P(A \cup B) \leq P(A) + P(B).$$

For each of these three inequalities, give a simple criterion for when the inequality is actually an equality (e.g., give a simple condition such that $P(A \cap B) = P(A \cup B)$ if and only if the condition holds).

Solution

According to the inclusion-exclusion formula,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

(i) Since $P(A \cup B) \leq 1$,

$$P(A) + P(B) - P(A \cap B) \leq 1 \implies P(A) + P(B) - 1 \leq P(A \cap B).$$

Equality holds if and only if $P(A \cup B) = 1$.

(ii) Since $A \cap B \subseteq A \cup B$,

$$P(A \cap B) \leq P(A \cup B).$$

Equality holds if and only if $P(A \Delta B) = 0$.

(iii) Since $P(A \cap B) \geq 0$,

$$P(A) + P(B) - P(A \cup B) \geq 0 \implies P(A \cup B) \leq P(A) + P(B).$$

Equality holds if and only if $P(A \cap B) = 0$.

Problem 2

Let A and B be events. The *symmetric difference* $A \triangle B$ is defined to be the set of all elements that are in A or B but not both. In logic and engineering, this event is also called the XOR (*exclusive or*) of A and B . Show that

$$P(A \triangle B) = P(A) + P(B) - 2P(A \cap B),$$

directly using the axioms of probability.

Solution

By definition,

$$A \triangle B = (A \setminus B) \cup (B \setminus A).$$

The events $A \setminus B$ and $B \setminus A$ are disjoint, so

$$P(A \triangle B) = P(A \setminus B) + P(B \setminus A).$$

As

$$A = (A \setminus B) \cup (A \cap B), \quad B = (B \setminus A) \cup (A \cap B),$$

where each union is disjoint, we have

$$P(A \setminus B) = P(A) - P(A \cap B), \quad P(B \setminus A) = P(B) - P(A \cap B).$$

Therefore,

$$\begin{aligned} P(A \triangle B) &= P(A) - P(A \cap B) + P(B) - P(A \cap B) \\ &= P(A) + P(B) - 2P(A \cap B). \end{aligned}$$

Problem 3

Events A and B are *independent* if $P(A \cap B) = P(A)P(B)$.

1. Give an example of independent events A and B in a finite sample space S (with neither equal to $\emptyset$ or S).
2. Consider the experiment of picking a random point in the rectangle

$$R = \{(x, y) : 0 < x < 1, 0 < y < 1\},$$

where the probability of the point being in any particular region contained within R is the area of that region. Let A_1 and B_1 be rectangles contained within R , with areas not equal to 0 or 1. Let A be the event that the random point is in A_1 , and B be the event that the random point is in B_1 . Give a geometric description of when it is true that A and B are independent. Also, give an example where they are independent and another example where they are not independent.

3. Show that if A and B are independent, then

$$P(A \cup B) = P(A) + P(B) - P(A)P(B) = 1 - P(A^c)P(B^c).$$

Solution

- (i) Assume $S = \{1, 2, 3, 4\}$, $A = \{1, 2\}$, and $B = \{2, 4\}$. Then

$$P(A) = \frac{1}{2}, \quad P(B) = \frac{1}{2}, \quad A \cap B = \{2\},$$

and

$$P(A \cap B) = \frac{1}{4} = P(A)P(B).$$

Thus A and B are independent.

- (ii) Let S_{A_1} , S_{B_1} , and $S_{A_1 \cap B_1}$ denote the corresponding areas. Then

$$P(A) = S_{A_1}, \quad P(B) = S_{B_1}, \quad P(A \cap B) = S_{A_1 \cap B_1}.$$

The events A and B are independent when

$$S_{A_1}S_{B_1} = S_{A_1 \cap B_1}.$$

Geometrically, the overlap of A_1 and B_1 should have an area equal to the product of their individual areas.

For an independent example, let

$$A_1 = \{(x, y) : 0 < x < \frac{1}{2}, 0 < y < 1\}, \quad B_1 = \{(x, y) : 0 < x < 1, 0 < y < \frac{1}{2}\}.$$

Then

$$P(A) = \frac{1}{2}, \quad P(B) = \frac{1}{2}, \quad P(A \cap B) = \frac{1}{4} = P(A)P(B).$$

For a non-independent example, let

$$A_1 = B_1 = \{(x, y) : 0 < x < \tfrac{1}{2}, 0 < y < 1\}.$$

Then

$$P(A) = P(B) = \tfrac{1}{2}, \quad P(A \cap B) = \tfrac{1}{2} \neq P(A)P(B).$$

(iii) Since A and B are independent, $P(A \cap B) = P(A)P(B)$. Thus

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = P(A) + P(B) - P(A)P(B).$$

Also,

$$P(A^c)P(B^c) = P((A \cup B)^c) = 1 - P(A \cup B),$$

so

$$P(A \cup B) = 1 - P(A^c)P(B^c).$$

Problem 4

Consider four nonstandard dice (the *Efron dice*), whose sides are labeled as follows (the 6 sides on each die are equally likely).

A: 4, 4, 4, 4, 0, 0

B: 3, 3, 3, 3, 3, 3

C: 6, 6, 2, 2, 2, 2

D: 5, 5, 5, 1, 1, 1

These four dice are each rolled once. Let A be the result for die A , B be the result for die B , etc.

- Find $P(A > B)$, $P(B > C)$, $P(C > D)$, and $P(D > A)$.
- Is the event $A > B$ independent of the event $B > C$? Is the event $B > C$ independent of the event $C > D$? Explain.

Solution

- (i) The event $A > B$ occurs when $A = 4$, so

$$P(A > B) = \frac{4}{6} = \frac{2}{3}.$$

The event $B > C$ occurs when $C = 2$, so

$$P(B > C) = \frac{4}{6} = \frac{2}{3}.$$

The event $C > D$ occurs when $C = 6$, or when $C = 2$ and $D = 1$. These cases have probabilities

$$p_1 = \frac{2}{6} = \frac{1}{3}, \quad p_2 = \frac{2}{3} \cdot \frac{1}{2} = \frac{1}{3}.$$

Hence

$$P(C > D) = \frac{1}{3} + \frac{1}{3} = \frac{2}{3}.$$

The event $D > A$ occurs when $D = 5$, or when $D = 1$ and $A = 0$. These cases have probabilities

$$p_1 = \frac{1}{2}, \quad p_2 = \frac{1}{2} \cdot \frac{1}{3} = \frac{1}{6}.$$

Hence

$$P(D > A) = \frac{1}{2} + \frac{1}{6} = \frac{2}{3}.$$

- (ii) We have $P(A > B) = P(B > C) = \frac{2}{3}$. The event $\{A > B\} \cap \{B > C\}$ occurs when $A = 4$, $B = 3$, and $C = 2$. Thus

$$P(\{A > B\} \cap \{B > C\}) = \frac{2}{3} \cdot \frac{1}{3} \cdot \frac{2}{3} = \frac{4}{9} = P(A > B)P(B > C).$$

Therefore, the events $A > B$ and $B > C$ are independent.

We also have $P(B > C) = P(C > D) = \frac{2}{3}$. The event $\{B > C\} \cap \{C > D\}$ occurs when $B = 3$, $C = 2$, and $D = 1$. Thus

$$P(\{B > C\} \cap \{C > D\}) = 1 \cdot \frac{2}{3} \cdot \frac{1}{2} = \frac{1}{3} \neq P(B > C)P(C > D).$$

Therefore, the events $B > C$ and $C > D$ are not independent, because $B > C$ implies $C = 2$, which changes the probability that $C > D$.